UCCN 2003
UCCN 2243

TCP/IP Internetworking, Internetwork Principles & Practices

(Lecture 04a)

IPv6 Fundamentals: IPv6 Basic Rules

Short Stories of IPv*X*

IPv4 to IPv6, but where is IPv5?

- In the late 1970's, a protocol named ST (Stream Transport) — The Internet Stream Protocol — was created for the experimental transmission of voice, video, and distributed simulation over the Internet.
- This ST protocol was later given a name of IP Version 5 (IPv5)
  - IPv5 was intended as a IP-layer protocol that provides end-to-end guaranteed service across a network.
  - That is, it was compatible with IP at the network layer but was built to provide a Quality of Service for streaming services.
  - IPv5 gained some following at places like IBM, Apple, and Sun, but never really saw the light of day.
- In short, IPv5 never became an official protocol, because the work of developing ST (IPv5) was abandoned before ever becoming a standard.
- Regardless of its popularity, this ST protocol was already given the designation IPv5 and as a result, the next generation Internet protocol (after IPv4) couldn't take the name IPv5 and is thus called IPv6.
- The design and development of Internet Protocol Version 6 (IPv6), was initiated by the Internet Engineering Task Force (IETF) as early as in 1994.

What Happened to IPv0, v1, v2, v3,...?

0	IP	March 1977 version	(deprecated)
1	IP	January 1978 version	(deprecated)
2	IP	February 1978 version A	(deprecated)
3	IP	February 1978 version B	(deprecated)
4	**IPv4**	**September 1981 version**	**(current widespread)**
5	ST	Stream Transport	(not a new IP, little use)
6	**IPv6**	**December 1998 version**	**(formerly SIP, SIPP)**
7	CATNIP	IPng evaluation	(formerly TP/IX; deprecated)
8	Pip	IPng evaluation	(deprecated)
9	TUBA	IPng evaluation	(deprecated)
10-15		unassigned	

The term deprecation is applied to software features that are superseded and should be avoided.

IPng = IP next generation

A lot of IPvx only remained in the lab and proposal stage….

Other versions of IP…

- IPv4 and ST were both specified in the late 1970's.
- ST was a complementary protocol to IP, providing a connection-oriented internetwork service that was used to support early experiments in Internet audio/video teleconferencing.
- SIP[P], CATNIP, Pip, and TUBA were the four candidates considered in the early 1990's by the IETF for the new IP.
  - SIPP was the "winner", after it was changed from having 64-bit addresses to 128-bit addresses.
- SIP = Simple Internet Protocol
- SIPP = Simple Internet Protocol Plus
- CATNIP = Common Address Technology for Next-Generation IP
- Pip = The P Internet Protocol
- TUBA = TCP and UDP with Bigger Addresses
  - aka CLNP, the OSI Connectionless Network Layer Protocol

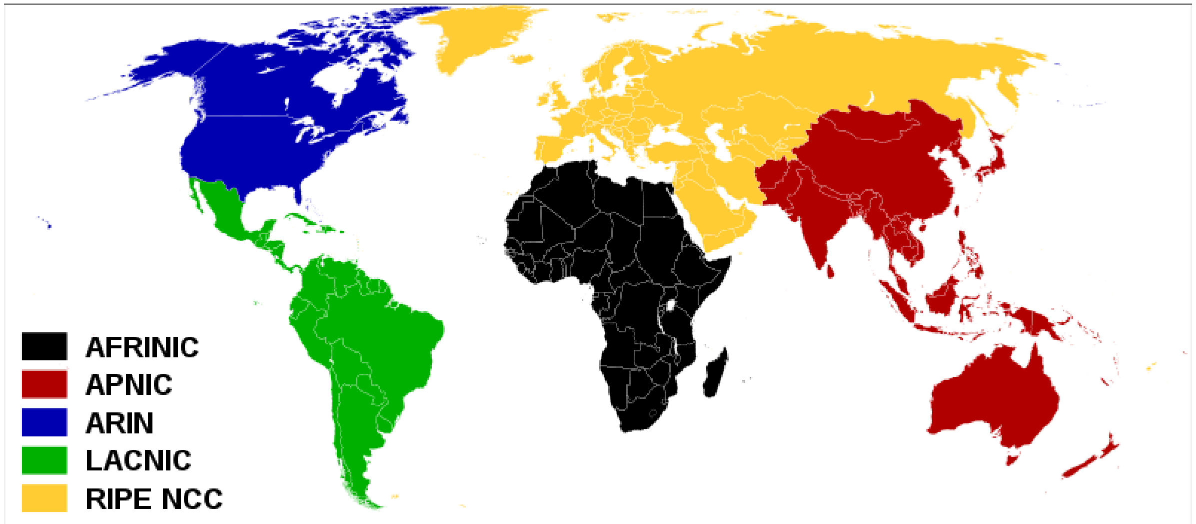
Motivation of Implementing IPv6 NOW!

IPv4 => NAT => IPv6

- IPv4 addresses have been projected to be in the final stages of depletion by 2008
  - Even with the help of NAT, NOT ENOUGH IPv4 addresses
  - IANA Unallocated Address Pool Exhaustion:
    - 01-Feb-2011
  - Projected RIR (Regional Internet registries) Unallocated Address Pool Exhaustion:
    - 09-Aug-2011
- IPv6 finally has to be pushed out to solve the exhaustion problem
  - Not much choice now….
  - Standard has been in place since 1994, and finalized back in 1998, but “resistance” to use IPv6 has been prevailing

5 RIR in the world

- There are five RIRs (regional Internet registry):
  - African Network Information Centre (AfriNIC) for Africa
  - American Registry for Internet Numbers (ARIN) for the United States, Canada, and several parts of the Caribbean region.
  - Asia-Pacific Network Information Centre (APNIC) for Asia, Australia, New Zealand, and neighboring countries
  - Latin America and Caribbean Network Information Centre (LACNIC) for Latin America and parts of the Caribbean region
  - RIPE NCC for Europe, the Middle East, and Central Asia



What is RIR then?

- The Internet Assigned Numbers Authority (IANA) delegates Internet resources to the RIRs who, in turn, follow their regional policies to delegate resources to their customers, which include Internet service providers and end-user organizations.

- A regional Internet registry (RIR)
  - is an organization that manages the allocation and registration of Internet number resources within a particular region of the world.
  - Internet number resources include IP addresses and autonomous system (AS) numbers.

Projected IPv4 Exhaustion

- /8s (or 24 bits) = 16,777,216 addresses.
  - APNIC 1.1981 => 16,777,216 * 1.1981
  - About 20100782 public IPv4 left in Asia-Pacific region.

IPv4 Address Report
This report generated at 07-Feb-2017 08:20 UTC.

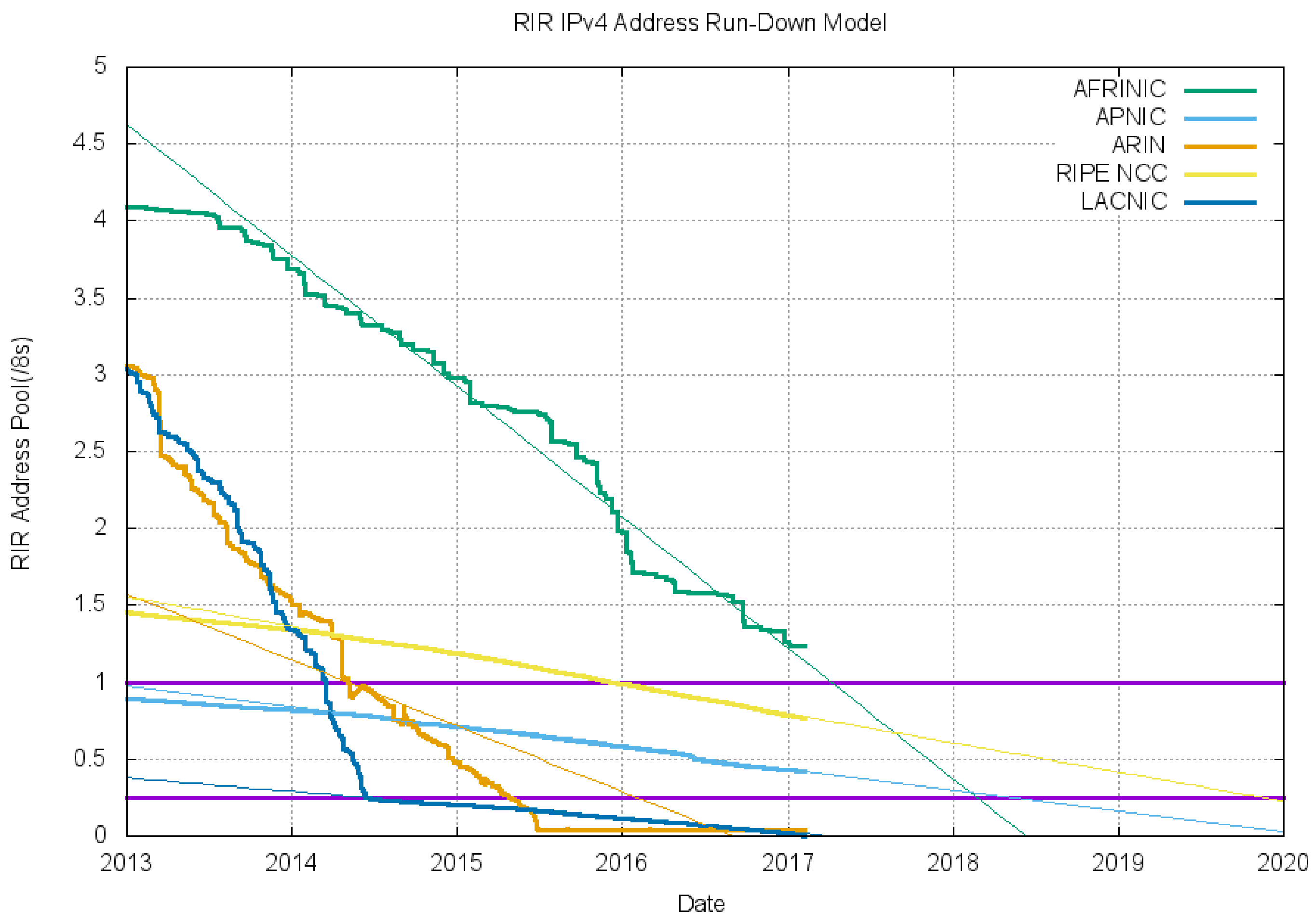
IANA Unallocated Address Pool Exhaustion:
03-Feb-2011

Projected RIR Address Pool Exhaustion Dates:

RIR	Projected Exhaustion Date	Remaining Addresses in RIR Pool (/8s)
APNIC:	**19-Apr-2011** (actual)	0.4203
RIPE NCC:	**14-Sep-2012** (actual)	0.7674
LACNIC:	**10-Jun-2014** (actual)	0.0036
ARIN:	**24 Sep-2015** (actual)	
AFRINIC:	**24-Jul-2018**	1.2336

Exhaustion" is defined here as the time when the pool of available addresses in each RIR reaches the "last /8 threshold" of 16,777,216 addresses.

Graphical Projection of IPv4 Exhaustation



Will increased use of NATs be adequate for IPv4?

NO!

- NAT enforces a ‘client-server’ application model where the server has topological constraints.
  - They won’t work for peer-to-peer or devices that are “called” by others (e.g., IP phones)
  - They inhibit deployment of new applications and services, because all NATs in the path have to be upgraded BEFORE the application can be deployed.
- NAT compromises the performance and robustness of the Internet.
- NAT increases complexity and reduces manageability of the local network.
- Public address consumption is still rising even with current NAT deployments.

Issues of NAT

- Advantages:
  - Reduce the need of official addresses
  - Ease the internal addressing plan
  - Transparent to some applications
  - “Security” in the form of masquerading
- Disadvantages:
  - Translation sometimes complex (e.g. FTP)
  - Apps using dynamic ports
  - Does not scale
  - Breaks the end-to-end paradigm
  - Security issues with IPsec

NAT implications

- Breaks end-to-end network model
  - Some applications cannot work through NATs
  - Breaks end-end security (IPsec)
- Requires application-level gateway (ALG)
  - When new application is not NAT-aware, ALG device must be upgraded
- Merging of separate private networks is difficult
  - Due to address clashes

IPv6 motivation

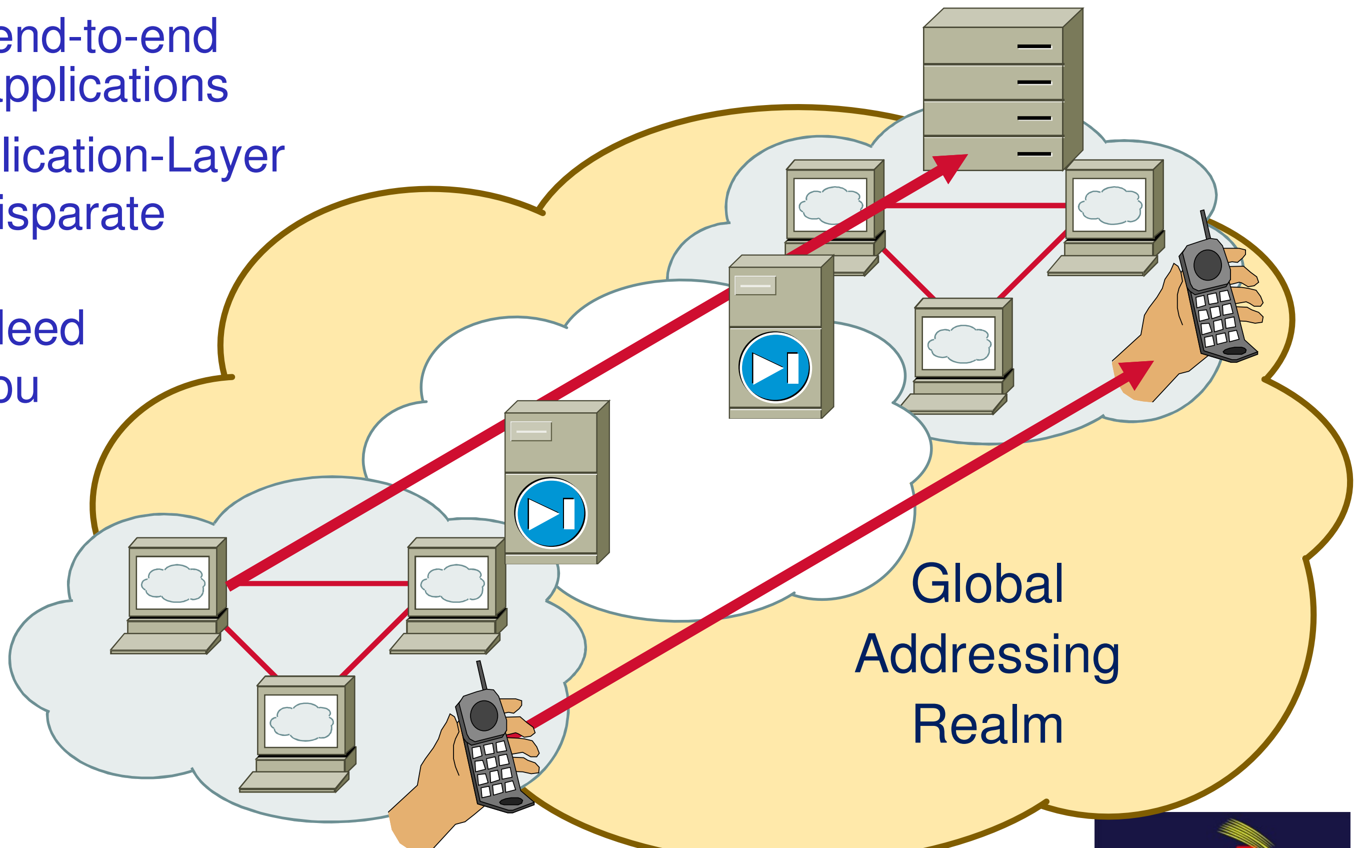
- The architecture of IPv6 has been designed to allow existing IPv4 users to transition easily to IPv6 while
  - providing services such as end-to-end security, quality of service (QoS),
  - providing globally unique addresses.
  - providing larger IPv6 address space allows networks to scale and provide global reachability.
  - simplified IPv6 packet header format handles packets more efficiently.

We need to go back to an End-to-End Secured Architecture

New Technologies/Applications for Home Users

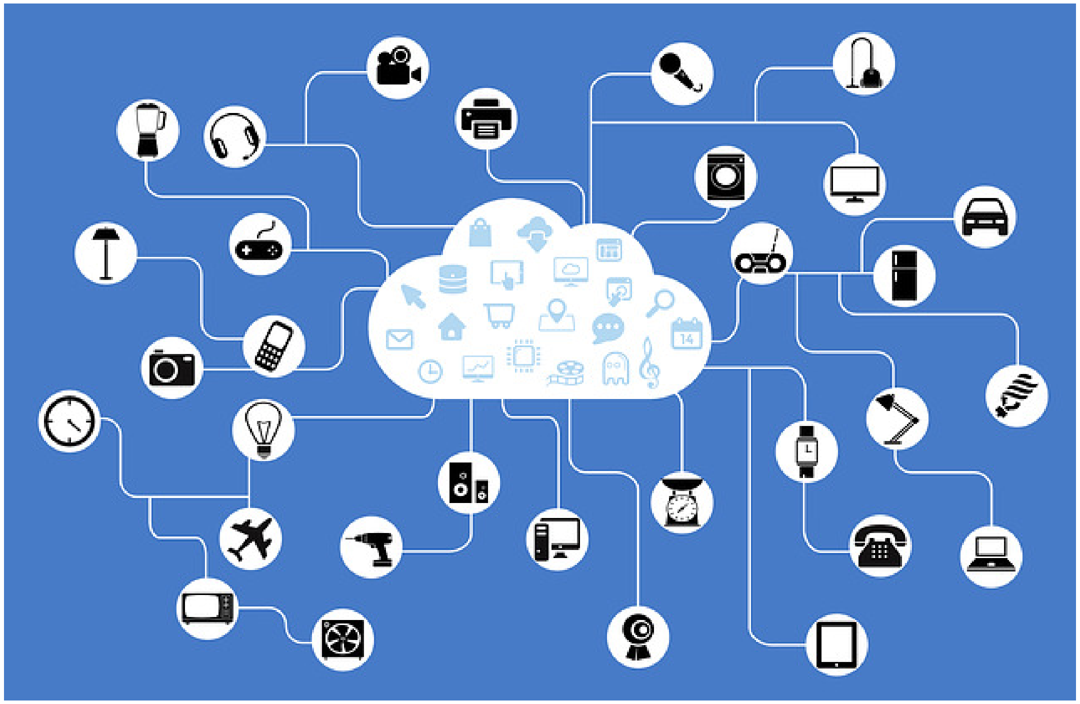
‘Always-on’—Cable, DSL, Ethernet-to-the-home, Wireless,…

- Internet started with end-to-end connectivity for any applications
- Today, NAT and Application-Layer Gateways connect disparate networks
- Always-on Devices Need an Address When You Call Them, eg.
  - Mobile Phones
  - Gaming
  - Residential Voice over IP gateway
  - IP Fax



IPv6 Strengths

- Larger Addresses
  - Allows billions of devices to be interconnected
    - *Especially mobile devices*



Source: https://pixabay.com/en/network-iot-internet-of-things-782707/

IPv6 Strengths

- Larger Address pool means no NAT in many future deployment scenarios
  - Eliminate NAT architectures as a means of address scaling
  - Allow coherent end-to-end packet delivery
  - Improve the potential for use of end-to-end security tools for encryption and authentication
  - the flexibility of the IPv6 address space reduces the need for private addresses
  - Allow for widespread deployment peer-to-peer applications
    - SIP, IMM, …

- No NATS
  - Regain fundamental leverage of IP as a network architecture
    - Simple interior service requirement
    - Service environment defined as end-to-end application
  - This is a VERY GOOD THING
    - Complex network architectures scale very poorly!
    - Simple architectures will service a Giganet

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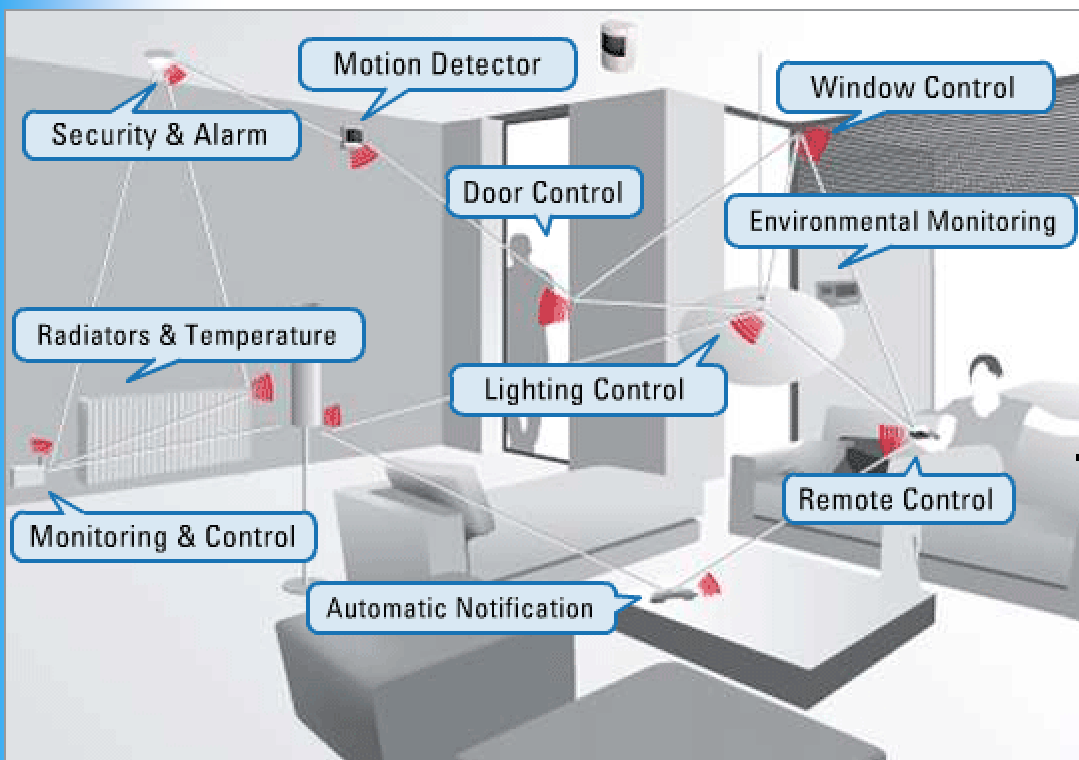
Imagine…..

- Next time, you will control your smart home (where every device is networked) with your smart phone via the Internet.
  - If there is no end-to-end secured connection….



IPv6 Address & Subnet Rules

Some Terminology

node	a protocol module that implements IPv6
router	a node that forwards IPv6 packets not explicitly addressed to itself
host	any node that is not a router
link	a communication facility or medium over which nodes can communicate at the link layer, i.e., the layer immediately below IPv6
neighbors	nodes attached to the same link
interface	a node's attachment to a link
address	an IPv6-layer identifier for an interface or a set of interfaces

IPv6 Address & Subnet Rule #1

IPv6 Addressing Space

IPv6 Rule #1

- IPv6 address is designed to be 128 bits long after much debate.

IPv4 32-bits

IPv6 128-bits

2^{32} = 4,294,967,296

2^{128} = 340,282,366,920,938,463,463,374,607,431,768,211,456

$2^{128} = 2^{32} * 2^{96}$

2^{96} = 79,228,162,514,264,337,593,543,950,336 times the number of possible IPv4 Addresses (79 trillion trillion)

Comparison of IPv4 and IPv6

IPv4:	4 octets
11000000.10101000.11001001.0111000	
192.168.201.113	
4,294,467,295 IP addresses	

- Currently, there are approximately 1.3 billion usable IPv4 addresses available.

IPv6:	16 octets
11010001.11011100.11001001.01110001.11010001.11011100. 11001100.01110001.11010001.11011100.11001001.01110001. 11010001.11011100.11001001.01110001	
A524:72D3:2C80:DD02:0029:EC7A:002B:EA73	
3.4 x 10^{38} IP addresses	

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Amount of IPv6 Address

World's population is approximately 6.5 billion

$$\frac{2^{128}}{6.5\ \text{Billion}}$$ = 52 Trillion Trillion IPv6 addresses per person

6.7 x 10^{23} Addresses / m^2 on the earth

Typical brain has ~100 billion brain cells (your count may vary)

$$\frac{52\ \text{Trillion Trillion}}{100\ \text{Billion}}$$ = 523 Quadrillion (523 thousand trillion) IPv6 addresses for every human brain cell on the planet!

IPv6 Address & Subnet Rule #2

IPv6 Address Format

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IPv6 Rule #2

2001:0DA8:E800:0000:0260:3EFF:FE47:0001

- IPv6 address format is represented in 32 hexadecimal numbers
  - IPv6 addresses are represented as a series of 16-bit hexadecimal fields separated by colons (:) in the format: x:x:x:x:x:x:x:x.
  - Each hexadecimal is 4 bits
    - 128 bits = 4 bits (1 hex digit) x 32
  - Written in 8 groups of 4 hexadecimal digits
    - Each group represents 16 bits
    - Separator is ":"

IPv6 Address & Subnet Rule #3

IPv6 Address Abbreviation
(Only for the zeros)

IPv6 Rule #3

- Two ways that an IPv6 address may be represented in a more abbreviated form.
  - However, the rule of IPv6 abbreviation only applies to string of zeros
- First, leading zeros in each 16-bit field may be omitted.
  - 2001:0000:0001:0002:0000:0000:0000:ABCD => 2001:0:1:2:0:0:0:ABCD
  - Leading zeros in a field are optional, so 09C0 = 9C0 and 0000 = 0.
  - One can NOT abbreviate the trailing zeros: 2001::<u>1200</u>:0:1

IPv6 Rule #3

- Secondly, once and only once, in an address, sequential zeros can be replaced with a pair of colons (::)
  - 2001:0:1:2:0:0:0:ABCD => 2001:0:1:2::ABCD
  - Successive fields of zeros can be represented as "::" only once in an address.
  - Using the "::" notation greatly reduces the size of most addresses. For example, FF01:0:0:0:0:0:0:1 becomes FF01::1.
  - Not valid!! 2001::123:A::2 (:: appear twice)
  - ::1 = loopback address
  - :: = unspecified address

Example of IPv6 Abbreviation

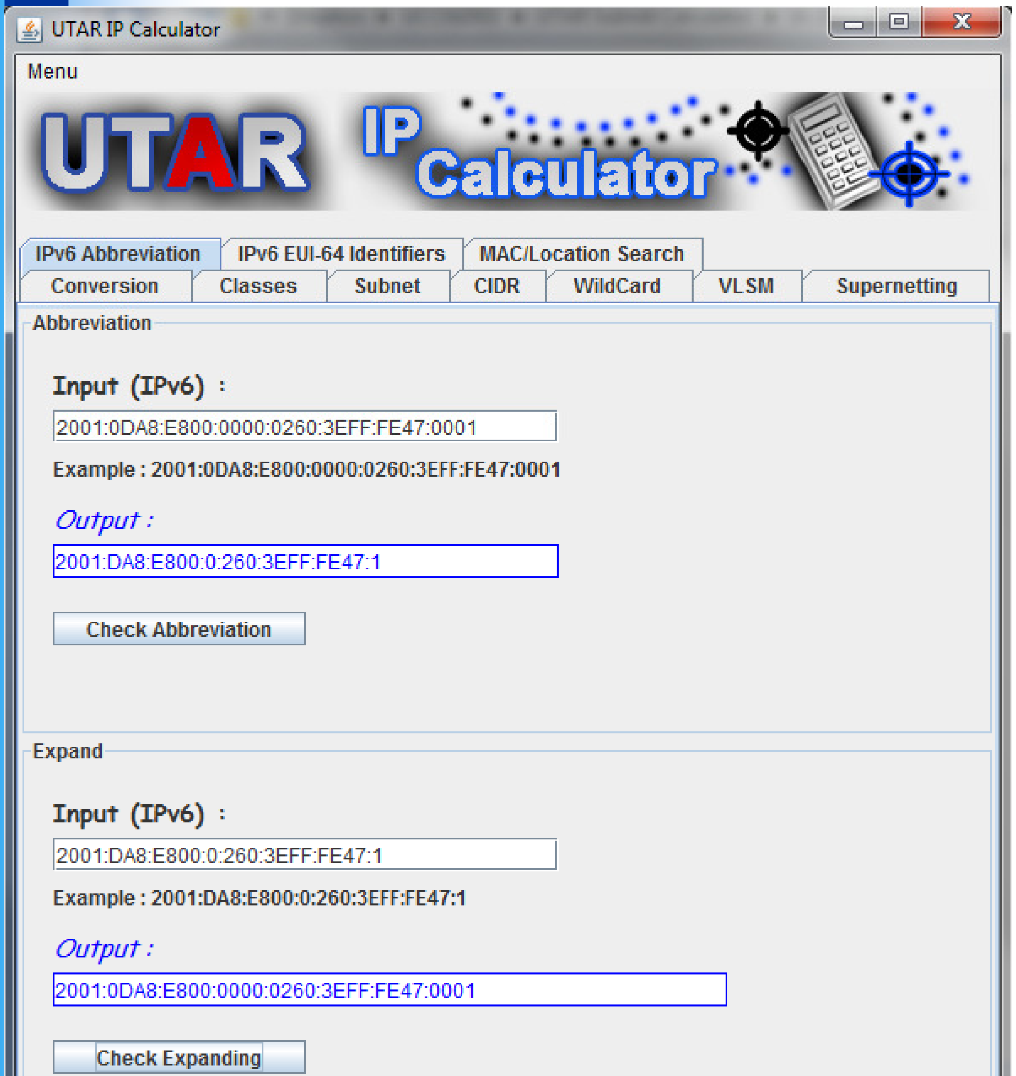
2001:0DA8:E800:0000:0260:3EFF:FE47:0001

2001:DA8:E800:0:260:3EFF:FE47:1

2001:0DA8:E800:0000:0000:0000:0000:0001

2001:DA8:E800::1

UTAR IP Calculator



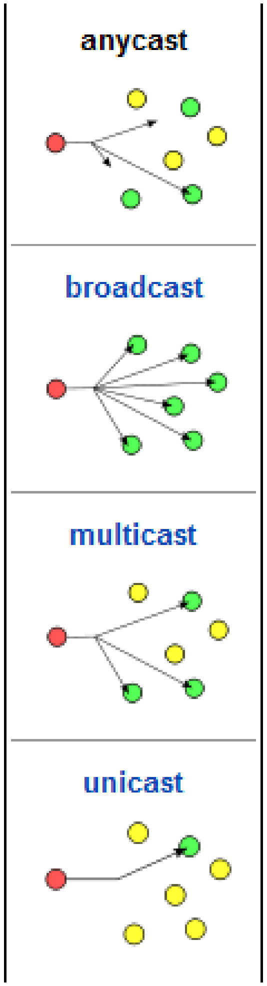
- You may use UTAR IP Calculator to help you on the IPv6 abbreviation.
- And also “expansion” based on an abbreviated IPv6 address.

IPv6 Address & Subnet Rule #4

Type of IPv6 Addresses:
Unicast, Multicast, Anycast

IPv6 Rule #4

- IPv6 supports three types of addresses:
  - Unicast—for sending to a single interface.
  - Multicast—for sending to all of the interfaces in a group or multiple interfaces.
  - Anycast—for sending to the nearest interface in a group.
- IPV6 does not support broadcast address.



IPv6 Address Types

- Unicast
  - Address is for a single interface.
  - One-to-one delivery to single interface.
  - IPv6 has several types (for example, global, reserved, and link-local).

- Multicast
  - One-to-many.
  - Enables more efficient use of the network.
  - Uses a larger address range.

- Anycast
  - One-to-nearest (allocated from unicast address space).
  - Multiple devices share the same address.
  - All anycast nodes should provide uniform service (e.g DNS).
  - Source devices send packets to anycast address.
  - Routers decide on closest device to reach that destination.

IPv6 Unicast Address

- A unicast address identifies a single device.
  - Work similarly as IPv4 class A, B, C addresses
  - A packet sent to a unicast address is delivered to the interface identified by that address.

- All interfaces are required to have at least one link-local unicast address.

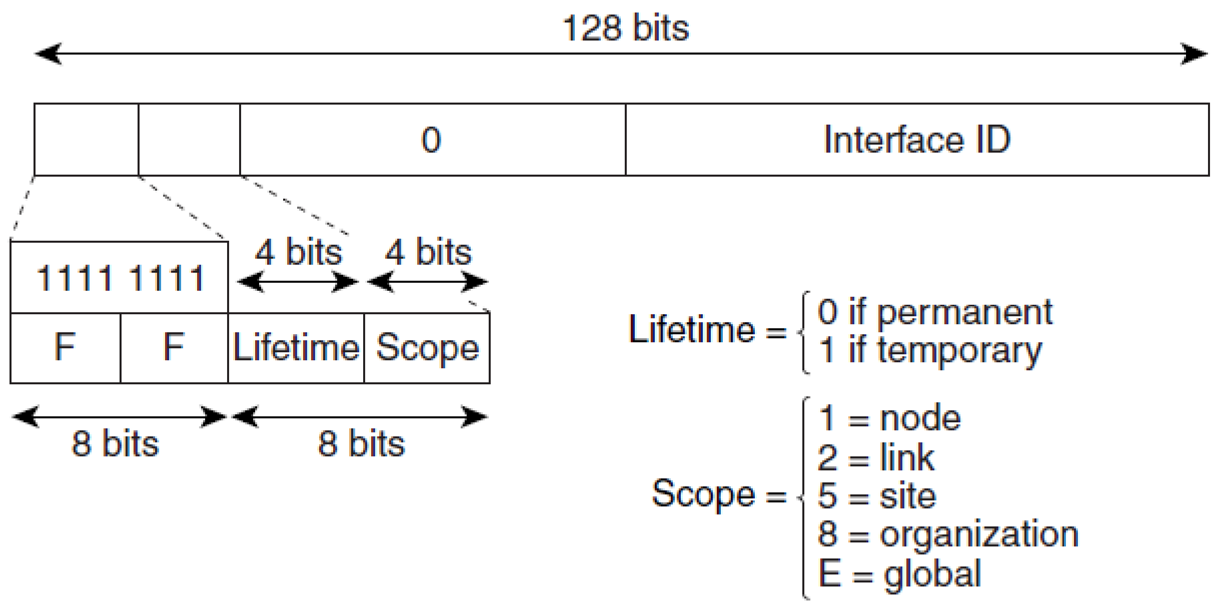
- There are a few types of IPv6 unicast addresses
  - To be elaborated in IPv6 subnet rule #5

IPv6 Multicast Address

- IPv6 broadcasts are replaced by multicast addresses.
- A packet sent to a multicast address is delivered to all interfaces identified by the multicast address.
- Multicast enables efficient network operation by using functionally specific multicast groups to send requests to a limited number of computers on the network.
- IPv6 multicast address work similarly as IPv4 multicast address.
- The range of multicast addresses in IPv6 is larger than in IPv4.

IPv6 Multicast Address

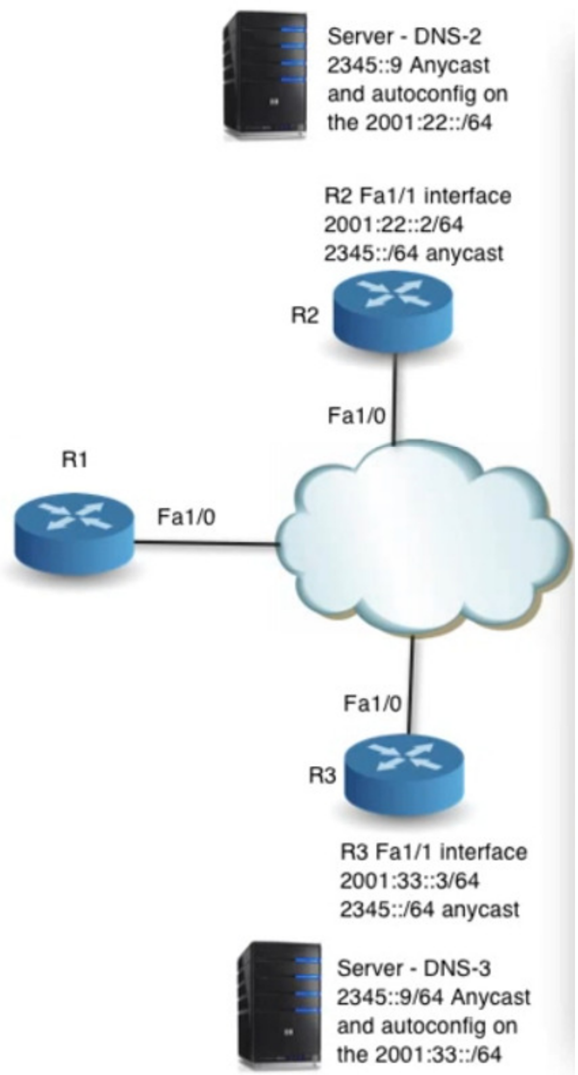
- A permanent multicast address has a lifetime parameter equal to 0; a temporary multicast address has a lifetime parameter equal to 1.
- A multicast address that has the scope of a node, link, site, or organization, or a global scope has a scope parameter of 1, 2, 5, 8, or E, respectively.
- For example, a multicast address with the prefix FF02::/16 is a permanent multicast address with a link scope.



Anycast Address

- An anycast address identifies a list of devices or nodes; therefore, an anycast address identifies multiple interfaces.
  - An IPv6 address that will be used more than once (or used on 2 or more devices).
- A packet sent to an anycast address is delivered to the closest interface, as defined by the routing protocols in use.
  - Route to the closest network, depending on the routing matric.
- Anycast addresses are syntactically indistinguishable from global unicast addresses, because anycast addresses are allocated from the global unicast address space.
- Anycast addresses cannot be used as the source address of an IPv6 packet.

Example of Anycast Usage



- Two DNS server having the same IPv6 "anycast" address.
  - DNS-2 and DNS-3
  - 2345::9
- If R1 wants to access the DNS, it just type 2345::9.
- Depending on the "nearest" route, R1 will access either 1 of the DNS server.

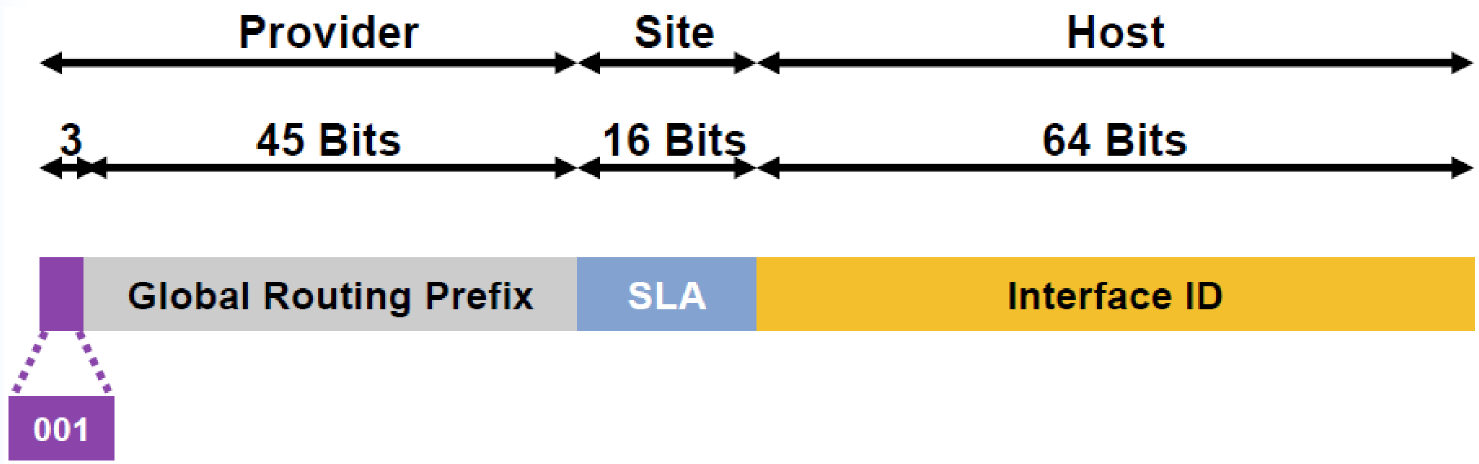
IPv6 Address & Subnet Rule #5

Type of IPv6 unicast address

IPv6 unicast address type

- An IPv6 unicast address is an identifier for a single interface, on a single node.
- A packet that is sent to a unicast address is delivered to the interface identified by that address.
- The Cisco IOS software supports the following IPv6 unicast address types:
  - Aggregatable Global Address
  - Link-Local Address
  - IPv4-Compatible IPv6 Address
  - Unique Local Address

Aggregatable Global Unicast Addresses

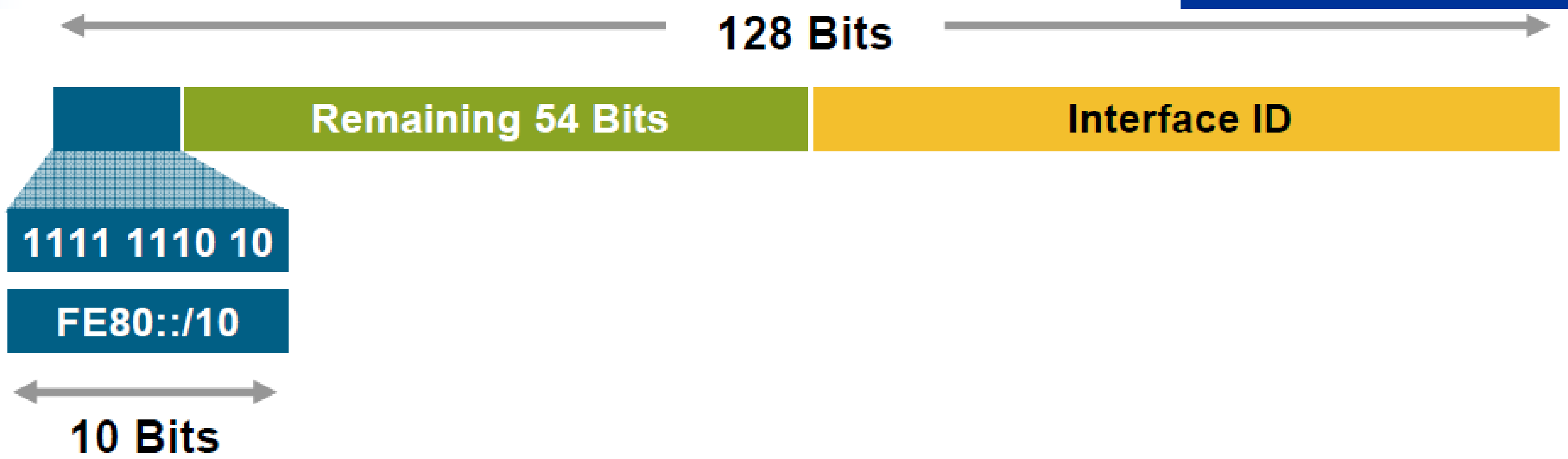


- Aggregatable Global Unicast Addresses Are:
  - Addresses for generic use of IPv6
  - Structured as a hierarchy to keep the aggregation

Aggregatable Global Unicast Addresses

- Aggregatable global IPv6 addresses are defined by a global routing prefix, a subnet ID, and an interface ID.
- Except for addresses that start with binary 000, all global unicast addresses have a 64-bit interface ID. The IPv6 global unicast address allocation uses the range of addresses that start with binary value 001 (2000::/3).
- The Internet Assigned Numbers Authority (IANA) allocates the IPv6 address space in the range of 2000::/16 to regional registries.
- A 16-bit subnet field called the subnet ID could be used by individual organizations to create their own local addressing hierarchy and to identify subnets.
- A subnet ID is similar to a subnet in IPv4, except that an organization with an IPv6 subnet ID can support up to 65,535 individual subnets.

Link-Local Addresses



- A link-local address is an IPv6 unicast address that can be automatically configured on any interface using the link-local prefix FE80::/10 (1111 1110 10) and the interface identifier in the modified EUI-64 format.

- Link-Local Addresses (similar to APIPA in IPv4) Used for:
  - Mandatory address for communication between two IPv6 device (like ARP but at layer 3)
  - Automatically assigned by router as soon as IPv6 is enabled
  - Also used for next-hop calculation in routing protocols
  - Only link-specific scope
  - Remaining 54 bits could be zero or any manual configured value

Link-Local Addresses

- Link-local addresses have a scope limited to the link and are dynamically created on all IPv6 interfaces by using a specific link-local prefix FE80::/10 and a 64-bit interface identifier.
- Link-local addresses are used for automatic address configuration, neighbor discovery, and router discovery.
- IPv6 routers must not forward packets that have link-local source or destination addresses to other links.
- Link-local addresses can serve as a way to connect devices on the same local network without needing global addresses.
- When communicating with a link-local address, you must specify the outgoing interface because every interface is connected to FE80::/10.

Example & Comments

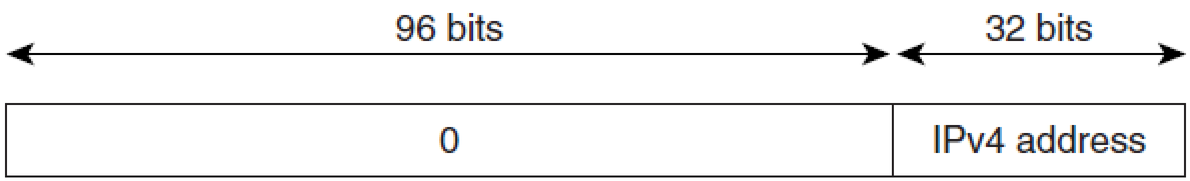
```
C:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 1:
        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 10.1.1.100
        Subnet Mask . . . . . . . . . . . : 255.255.255.0
        IP Address. . . . . . . . . . . . : 2001:DB8:C003:1122:203:ffff:fe81:d6da
        IP Address. . . . . . . . . . . . : fe80::203:ffff:fe81:d6da%4
        Default Gateway . . . . . . . . . : 10.1.1.1
                                            fe80::201:42ff:fe2d:9580
```

- Aggregatable global unicast addresses
  - You have to configure it.
- Local link addresses
  - Automatically formed by the OS

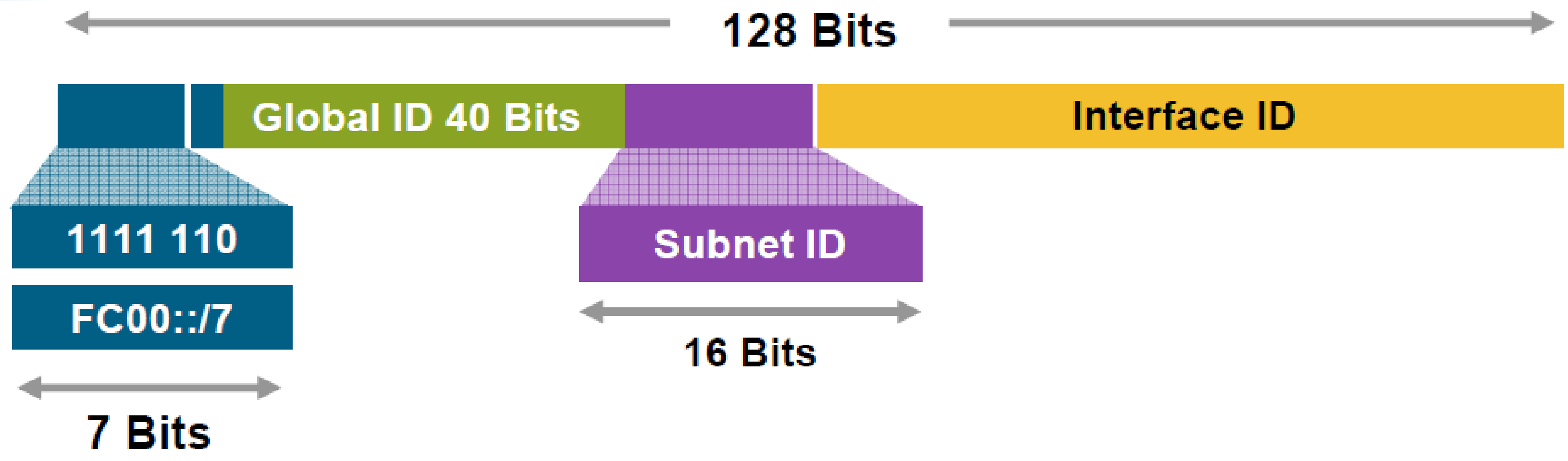
IPv4-Compatible IPv6 Address



::192.168.30.1
= ::C0A8:1E01

- An IPv4-compatible IPv6 address is an IPv6 unicast address that has zeros in the high-order 96 bits of the address and an IPv4 address in the low-order 32 bits of the address.
- The format of an IPv4-compatible IPv6 address is 0:0:0:0:0:0:A.B.C.D or ::A.B.C.D.
- The entire 128-bit IPv4-compatible IPv6 address is used as the IPv6 address of a node and the IPv4 address embedded in the low-order 32 bits is used as the IPv4 address of the node.
- IPv4-compatible IPv6 addresses are assigned to nodes that support both the IPv4 and IPv6 protocol stacks and are used in automatic tunnels.

Unique-Local Addresses



- Unique-Local Addresses (similar to Private IP address in IPv4) Used for:
  – Local communications
  – Inter-site VPNs
  – Not routable on the Internet

Unique-Local Addresses

- A unique local address is an IPv6 unicast address that is intended for local communications.
- They are not expected to be routable on the global Internet and are routable inside of a limited area, such as a site. They may also be routed between a limited set of sites.
- A unique local address has the following characteristics:
  - It has a globally unique prefix (that is, it has a high probability of uniqueness).
  - It has a well-known prefix to allow for easy filtering at site boundaries.
  - It allows sites to be combined or privately interconnected without creating any address conflicts or requiring renumbering of interfaces that use these prefixes.
  - It is ISP-independent and can be used for communications inside of a site without having any permanent or intermittent Internet connectivity.
  - If it is accidentally leaked outside of a site via routing or DNS, there is no conflict with any other addresses.
  - Applications may treat unique local addresses like global scoped addresses.

IPv6 Address & Subnet Rule #6

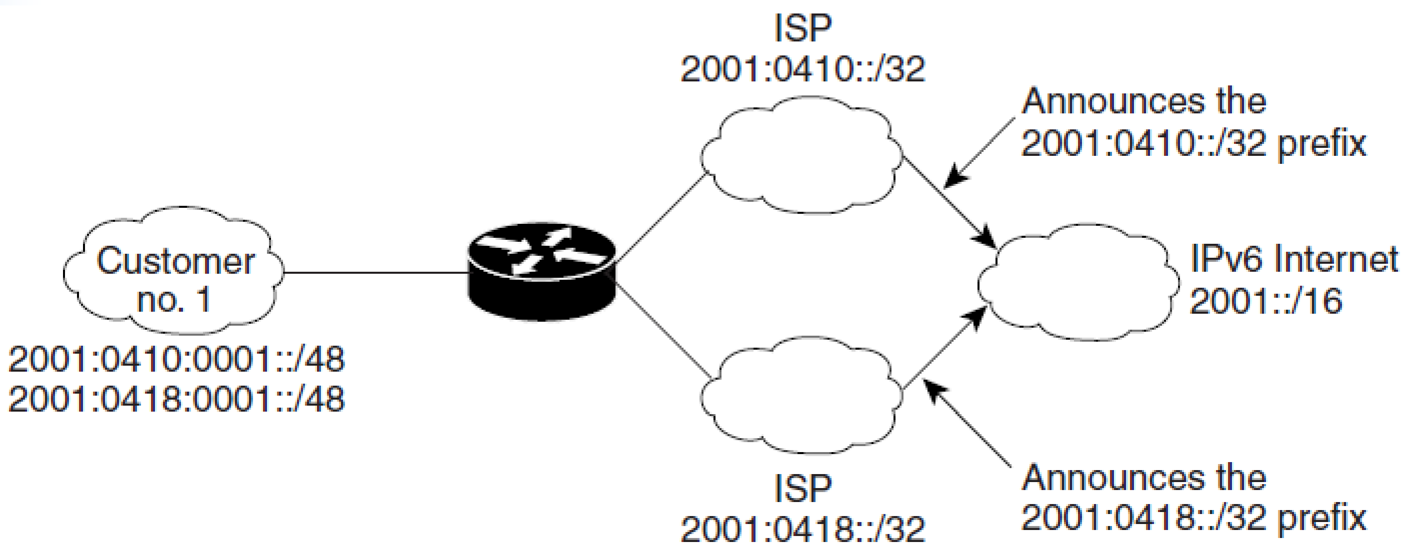
Multi-IPv6 Address in single interface
&
Multi-homing

IPv6 Rule #6 – Multi-IPv6 Addresses

- A fundamental feature of IPv6 is that a single interface may also have multiple IPv6 addresses of any type (unicast, anycast, and multicast).
- An interface can have several addresses, including a link-local address, any global unicast or anycast addresses assigned, a loopback address (::1/128), the all-nodes multicast addresses, solicited-nodes multicast addresses, and any other multicast addresses to which the node is assigned.
- In addition, routers must recognize the subnet-router anycast address and the all-routers multicast addresses.

IPv6 Site Multihoming

- Multiple IPv6 prefixes can be assigned to networks and hosts.
- Having multiple prefixes assigned to a network makes it easy for that network to connect to multiple ISPs without breaking the global routing table



IPv6 Multihoming

- Multihoming:
  - When a network is connected to more than one Internet service provider (ISP)
- The chief objective of multihoming
  - is to increase the quality and robustness of the Internet connection for the IP network.
  - It is also possible to extend this concept to devices, especially when each of them has more than one interface, and each of the interfaces is attached to different networks.

Multihoming Variant

- There are several ways to multihome:
  - Single Link, Multiple IP address
    - The host has multiple IP addresses (e.g. 2001:db8::1 and 2001:db8::2 in IPv6), but only one physical upstream link. When the single link fails, connectivity is down for all addresses.
  - Multiple Interfaces, Single IP address per interface (eg. obtain 2 independent connections to different ISPs)
    - The host has multiple interfaces and each interface has one IP address. If one of the links fails, then its IP address becomes unreachable, but the other IP addresses will still work.
  - Multiple Links, Single IP address (eg. obtain 2 connections to the same ISP with same IP address over both connections)
    - When one of the links fails, the protocol notices this on both sides and traffic is not sent over the failing link any more. Usually this method is used to multihome a site and not for single hosts.
  - Multiple Links, Multiple IP address (eg. obtain 2 connections from different ISPs but use the same range of IP address over both connections)
    - This approach uses a specialized Link Load Balancer (or WAN Load Balancer) appliance between the firewall and the link routers. No special configuration is required in the ISP's routers.

IPv6 Address & Subnet Rule #7

Special IPv6 Addresses

IPv6 Subnet Rule #7

- Address type identification
  - Unspecified 00..0 (128 bits) ::/128
  - Loopback 00..1 (128 bits) ::1/128
  - Link Local 1111 1110 10 FE80::/10
  - Multicast 1111 1111 FF00::/8
  - Global Unicast everything else

- All address types have to support EUI-64 bits Interface ID setting
  - Except for multicast

IPv6 Loopback & Unspecified Address

- Loopback address representation
  - 0:0:0:0:0:0:0:1=> ::1
  - Same as 127.0.0.1 in IPv4
  - Identifies self

- Unspecified address representation
  - 0:0:0:0:0:0:0:0=> ::
  - Used as a placeholder when no address available
  - (Initial DHCP request, Duplicate Address Detection, DAD)

IPv6 Address & Subnet Rule #8

IPv6 Subnet Mask

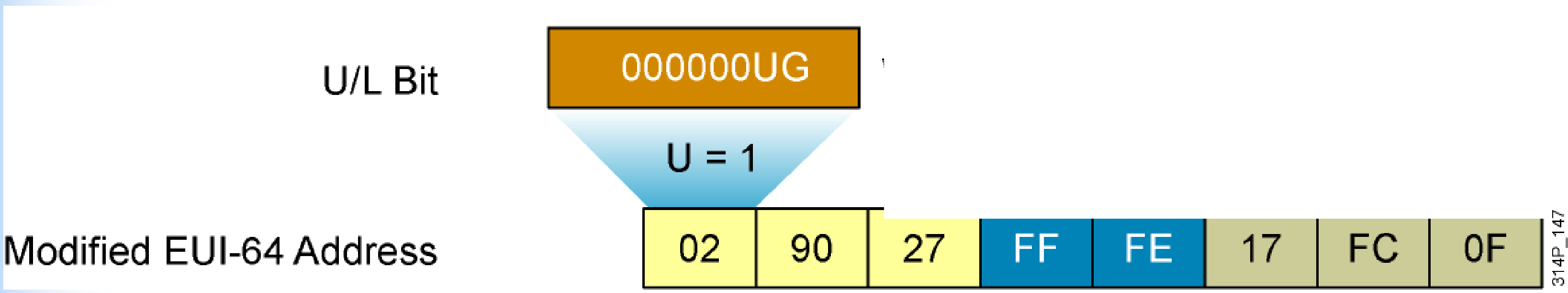
IPv6 Prefixes and Subnet Mask

- IPv6 “subnet mask” works the same as IPv4
  - But only slash notation ‘/’,
  - No 255.255.255……..
- Like v4 address:
  - 198.10.0.0/16
- IPv6 address is represented the same way:
  - 2001:db8:12::/48
- Prefixes = network ID.
- “Host ID” is always 64 bits.

IPv6 Address & Subnet Rule #9

Interface ID EUI-64

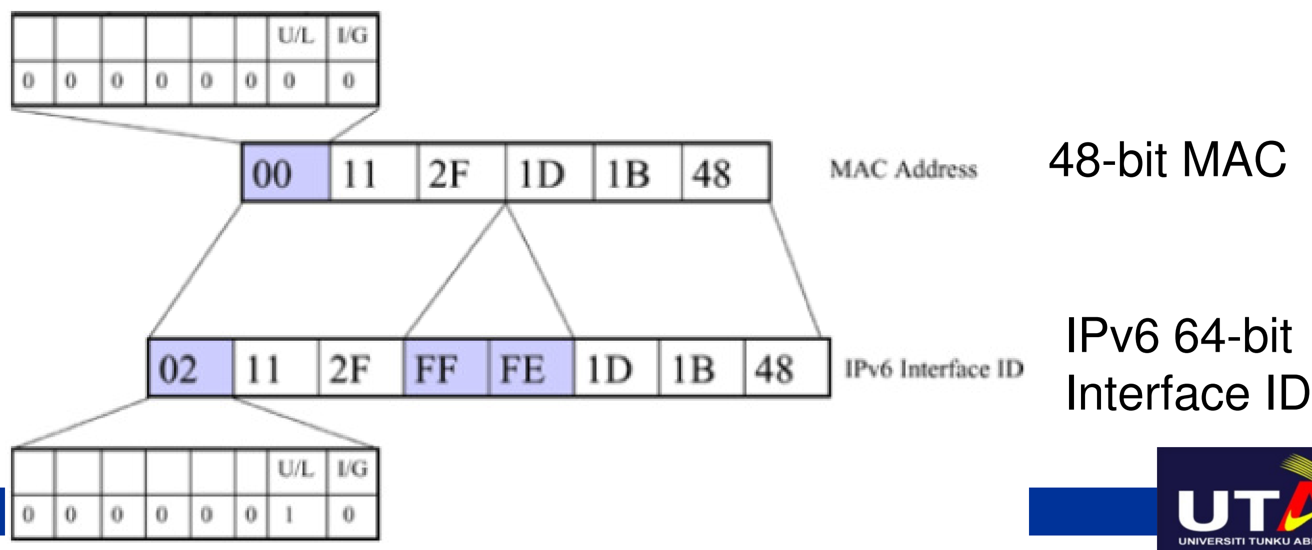
IPv6 EUI-64 Interface Identifier



- This format expands the 48-bit MAC address to 64 bits by inserting "FFFE" into the middle 16 bits.
- To indicate that the chosen address is from a unique Ethernet MAC address, the U/L bit is set to 1 for global scope (0 for local scope).

Interface ID

- The interface ID is based on the media access control (MAC) address of the interface, in a format called the extended universal identifier 64-bit (EUI-64) format.
- The EUI-64 format interface ID is derived from the 48-bit MAC address by inserting the hexadecimal number FFFE between the organizationally unique identifier (OUI) field (the upper three bytes) and the vendor code (the lower three bytes) of the MAC address.
- The seventh bit in the first byte of the resulting interface ID, corresponding to the Universal/Local (U/L) bit, is set to binary 1.



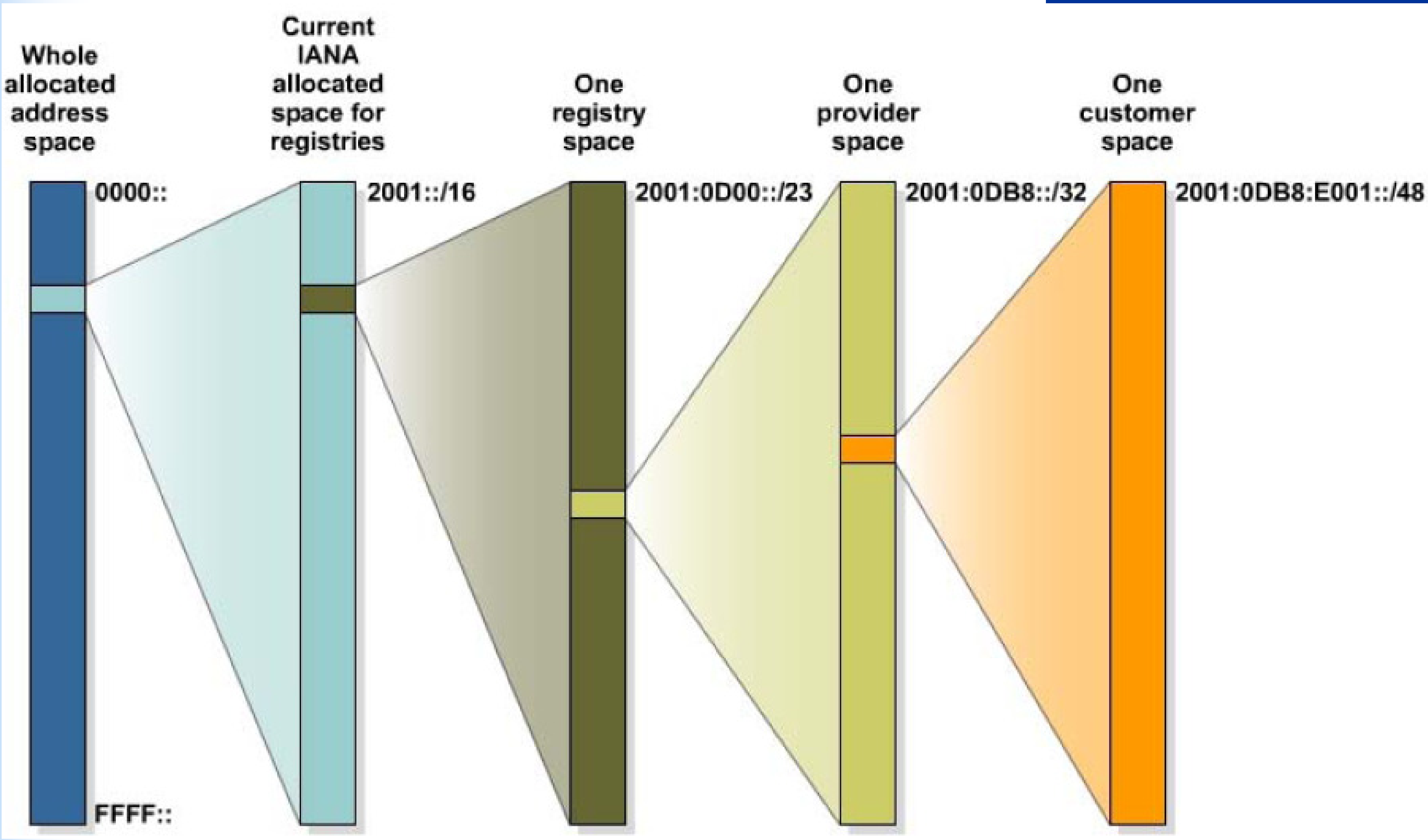
Example

- The IANA is currently allocating address allocate /32 ranges to ISPs.
- For example, an ISP might assign 2001:0:1AB::/48 to an organization.
- In a network assigned subnet 5, the prefix would be 2001:0:1AB:5::/64.
- On a device with a MAC address 00-0F-66-81-19-A3, the EUI-64 format interface ID would be 020F:66FF:FE81:19A3.
- The complete IPv6 global unicast address of the device would therefore be 2001:0:1AB:5:20F: 66FF:FE81:19A3.

IPv6 Address & Subnet Rule #10

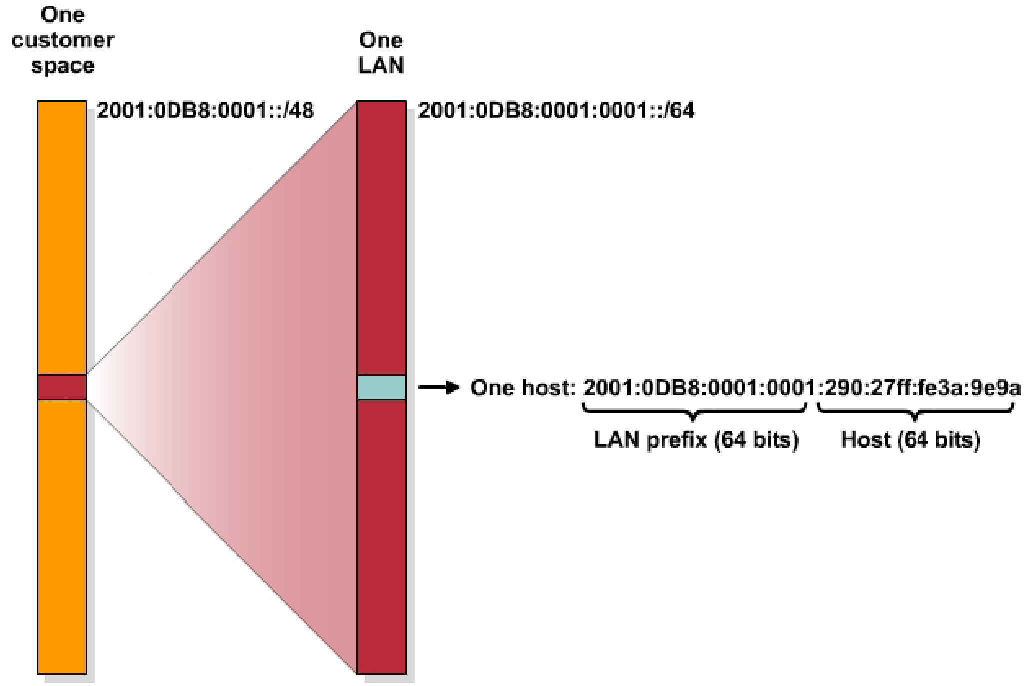
Prefix Hierarchy

IPv6 Address Allocation Process

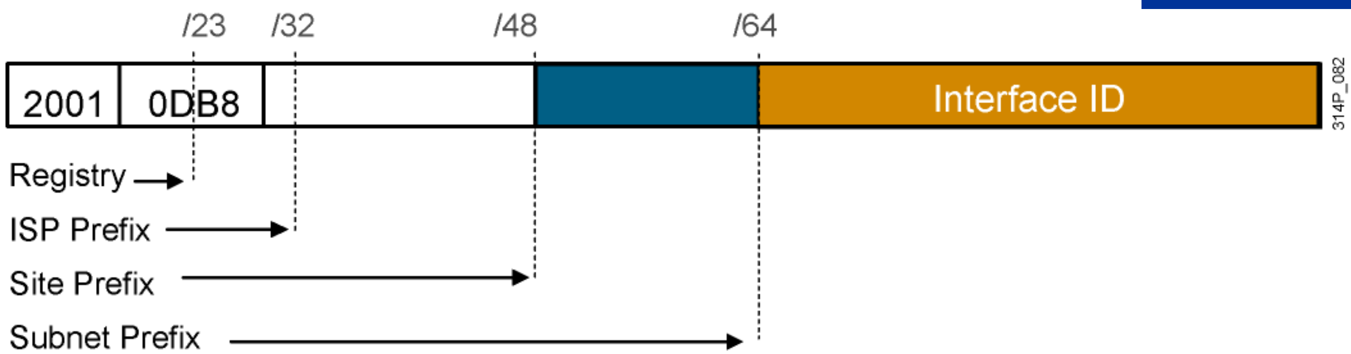


IPv6 Address Allocation Process

- Lowest-order 64-bit field of unicast address may be assigned in several different ways:
  - Autogenerated pseudo-random number (to address privacy concerns)
  - Assigned via DHCP
  - Manually configured
  - Auto-configured from a 64-Bit EUI-64, or expanded from a 48-bit MAC address (e.g., Ethernet address)



Assigning IPv6 Global Unicast Addresses



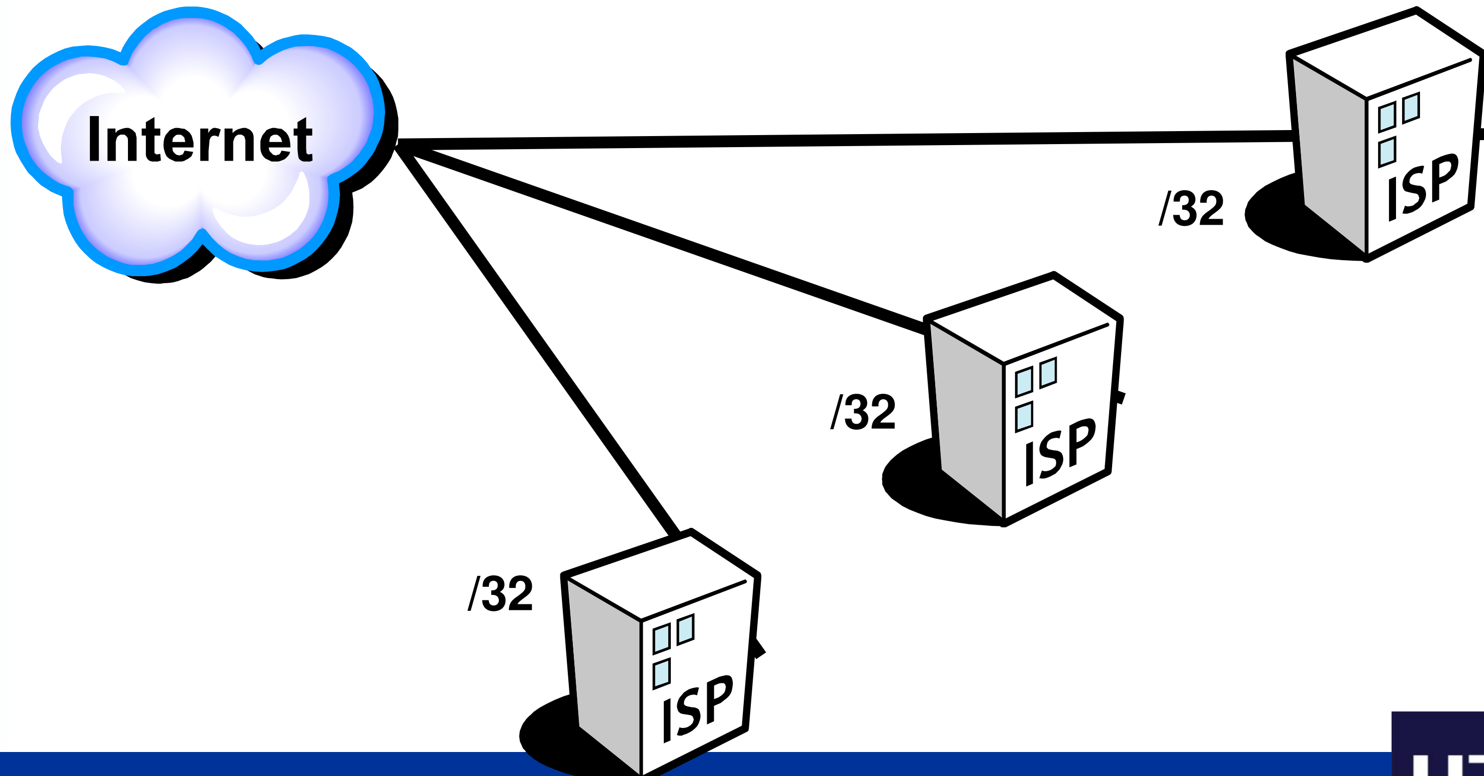
- Static assignment
  - Manual interface ID assignment
  - EUI-64 interface ID assignment
- Dynamic assignment
  - Stateless autoconfiguration
  - DHCPv6 (stateful)

```
RouterX(config-if) ipv6 address 2001:DB8:2222:7272::72/64
```

```
RouterX(config)# interface ethernet 0
RouterX(config-if)# ipv6 address 2001:0DB8:0:1::/64 eui-64
```

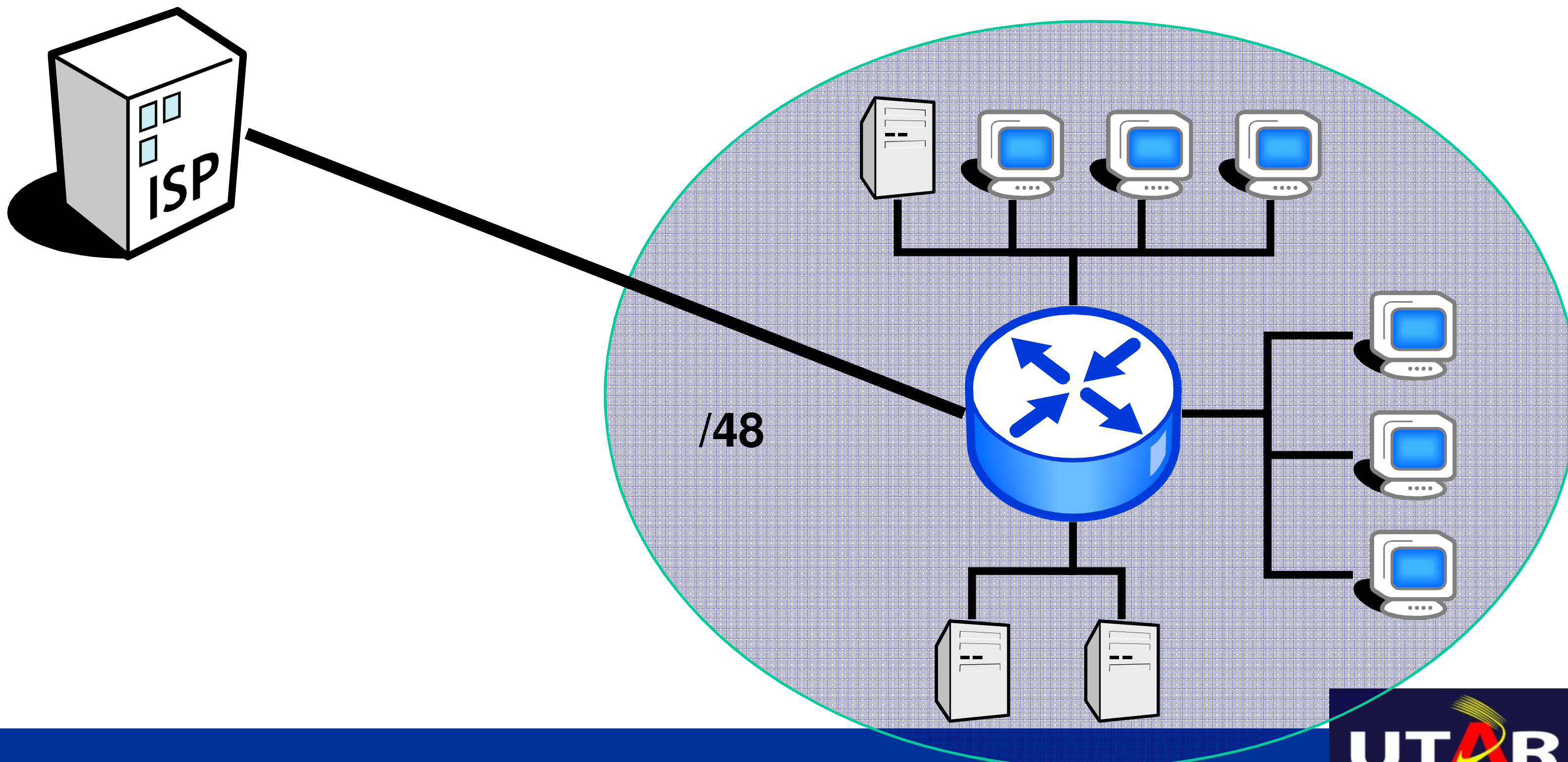
IPv6 – ISP addressing

- Every ISP receives a /32 (or more)
  - Providing 65,536 site addresses (/48)



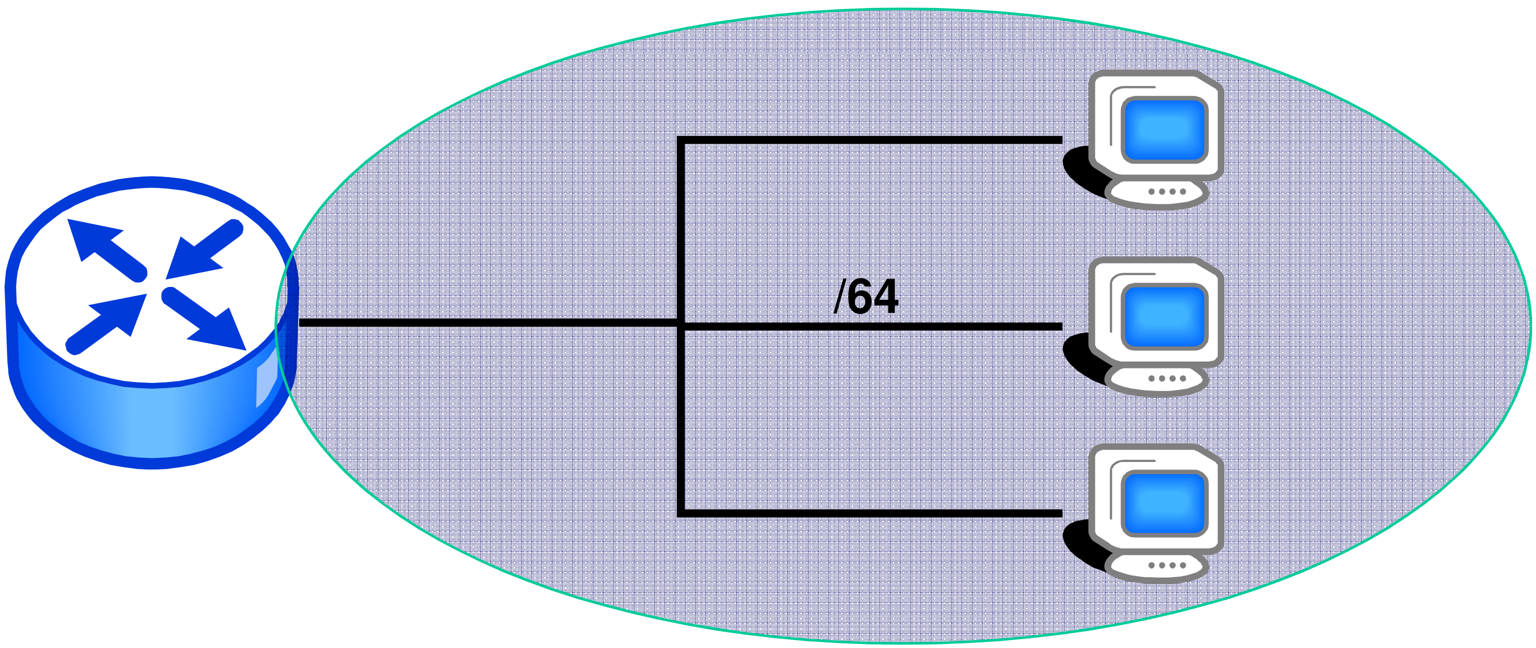
IPv6 – Site addressing

- Every "site" receives a /48
  - Providing 65,536 /64 (LAN) addresses



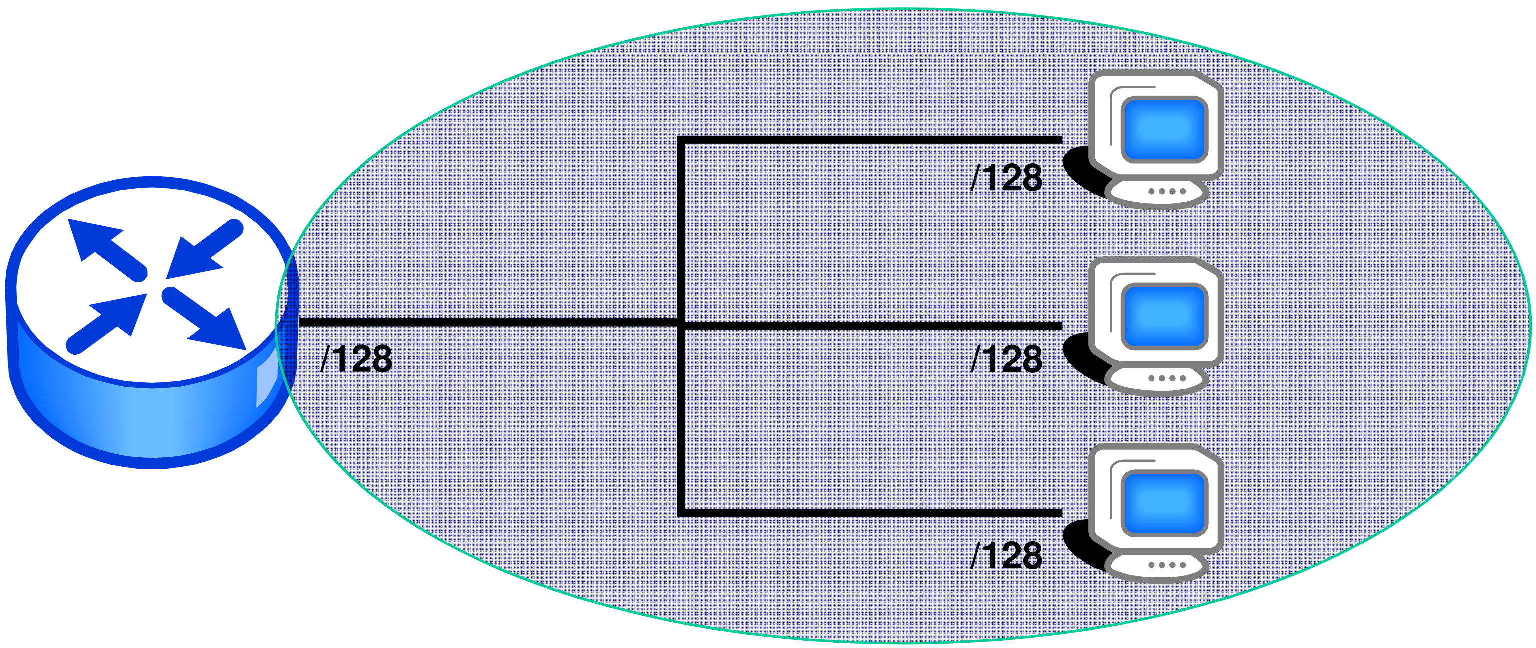
IPv6 – LAN addressing

- Every LAN segment receives a /64
  - Providing 2^{64} interface addresses per LAN



IPv6 – Device addressing

- Every device interface receives a /128
  - May be EUI-64 (derived from interface MAC address), random number (RFC 3041), autoconfiguration, or manual configuration



IPv6 Address & Subnet Rule #11

IPv6 Subnetting

IPv6 Subnetting

- IPv6 subnetting is greatly simplified, compared to IPv4.

- An IPv4 site address can be /8 till /24.
- A IPv6 site address is always /48

- IPv4 subnet bits => variable.
- IPv6 subnet bits => 16 bits.

- In IPv6 there is (usually, but NOT IMPOSSIBLE) no VLSM, classful subnetting, host number design…..
- In IPv6 host bits is always 64 bits.
- In IPv6 subnet bits is always 16 bits
  - Subnet #0 to Subnet #65,535

Introduction to Cisco IPv6 Configuration

IOS IPv6 Addressing Examples

Manual Interface Identifier

```
ipv6 unicast-routing
!
interface FastEthernet0/0
 ip address 10.151.1.1 255.255.255.0
 ip pim sparse-mode
 duplex auto
 speed auto
 ipv6 address 2006:1::1/64
 ipv6 enable
 ipv6 nd ra-interval 30
 ipv6 nd prefix 2006:1::/64 300 300
!
```

EUI-64 Interface Identifier

```
ipv6 unicast-routing
!
interface FastEthernet0/0
 ip address 10.151.1.1 255.255.255.0
 ip pim sparse-mode
 duplex auto
 speed auto
 ipv6 address 2006:1::/64 eui-64
 ipv6 enable
 ipv6 nd ra-interval 30
 ipv6 nd prefix 2006:1::/64 300 300
!
```

IOS IPv6 Addressing Examples

Manual Interface Identifier

```
r1#sh ipv6 int fast0/0
FastEthernet0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::207:50FF:FE5E:9460
  Global unicast address(es):
    2006:1::1, subnet is 2006:1::/64
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF00:1
    FF02::1:FF5E:9460
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ND
  ND
  ND
  ND advertised retransmit interval is 0 milliseconds
  ND router advertisements are sent every 30 seconds
  ND router advertisements live for 1800 seconds
  Hosts use stateless autoconfig for addresses.
r1#
```

MAC Address : 0007.505e.9460

```
r1#sh int fast0/0
FastEthernet0/0 is up, line protocol is up
  Hardware is AmdFE, address is 0007.505e.9460 (bia 0007.505e.9460)
```

IOS IPv6 Addressing Examples

EUI-64 Interface Identifier

```
r1#sh ipv6 int fast0/0
FastEthernet0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::207:50FF:FE5E:9460
  Global unicast address(es):
    2006:1::207:50FF:FE5E:9460, subnet is 2006:1::/64
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF5E:9460
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICM
  ND
  ND
  ND advertised reachable time is 0 milliseconds
  ND advertised retransmit interval is 0 milliseconds
  ND router advertisements are sent every 30 seconds
  ND router advertisements live for 1800 seconds
  Hosts use stateless autoconfig for addresses.
r1#
```

MAC Address : 0007.505e.9460

```
r1#sh int fast0/0
FastEthernet0/0 is up, line protocol is up
  Hardware is AmdFE, address is 0007.505e.9460 (bia 0007.505e.9460)
```

Multiple IPv6 Addresses on 1 Interface

- You can configure multiple IPv6 addresses onto 1 single interface.
  - But you can only configure one IPv4 address to a interface

To support IPv6 multihoming

```
Router(config)#ipv6 unicast
Router(config)#int fa0/0
Router(config-if)#ipv6 enable
Router(config-if)#ipv6 addr 2006:1::1/64
Router(config-if)#ipv6 addr 2006:1::/64 eui-64
Router(config-if)#ipv6 addr 2006:1::1000/64
Router(config-if)#ipv6 addr 2006:1::2002/64
Router(config-if)#no shut
```

```
Router#show ipv6 int br
FastEthernet0/0              [up/down]
    FE80::230:A3FF:FE20:201
    2006:1::1
    2006:1::1000
    2006:1::2002
    2006:1::230:A3FF:FE20:201
FastEthernet0/1              [administratively down/down]
Vlan1                        [administratively down/down]
```

Configuring IPv6 Addressing and Enabling IPv6 Routing for Cisco

SUMMARY STEPS

1. **enable**
2. **configure terminal**
3. **interface** *type number*
4. **ipv6 address** *ipv6-prefix/prefix-length* **eui-64**
   or
   ipv6 address *ipv6-address/prefix-length* **link-local**
   or
   ipv6 address *ipv6-prefix/prefix-length* **anycast**
   or
   ipv6 enable
5. **exit**
6. **ipv6 unicast-routing**

Detail Steps - 1

	Command or Action	Purpose
Step 1	**enable** **Example:** `Router> enable`	Enables privileged EXEC mode. • Enter your password if prompted.
Step 2	**configure terminal** **Example:** `Router# configure terminal`	Enters global configuration mode.
Step 3	**interface** *type number* **Example:** `Router(config)# interface ethernet 0/0`	Specifies an interface type and number, and places the router in interface configuration mode.

Detail Steps - 2

Step 4	**ipv6 address** *ipv6-prefix/prefix-length* **eui-64**
	or
	ipv6 address *ipv6-address/prefix-length* **link-local**
	or
	ipv6 address *ipv6-prefix/prefix-length* **anycast**
	or
	ipv6 enable
	
	Example:
	`Router(config-if)# ipv6 address 2001:0DB8:0:1::/64 eui-64`
	or
	
	Example:
	`Router(config-if)# ipv6 address FE80::260:3EFF:FE11:6770 link-local`
	or
	
	Example:
	`Router(config-if) ipv6 address 2001:0DB8:1:1:FFFF:FFFF:FFFF:FFFE/64 anycast`
	or
	
	Example:
	`Router(config-if)# ipv6 enable`
		Specifies an IPv6 network assigned to the interface and enables IPv6 processing on the interface.
		or
		Specifies an IPv6 address assigned to the interface and enables IPv6 processing on the interface.
		or
		Automatically configures an IPv6 link-local address on the interface while also enabling the interface for IPv6 processing. The link-local address can be used only to communicate with nodes on the same link.
		• Specifying the **ipv6 address eui-64** command configures global IPv6 addresses with an interface identifier (ID) in the low-order 64 bits of the IPv6 address. Only the 64-bit network prefix for the address needs to be specified; the last 64 bits are automatically computed from the interface ID.
		• Specifying the **ipv6 address link-local** command configures a link-local address on the interface that is used instead of the link-local address that is automatically configured when IPv6 is enabled on the interface.
		• Specifying the **ipv6 address anycast** command adds an IPv6 anycast address.

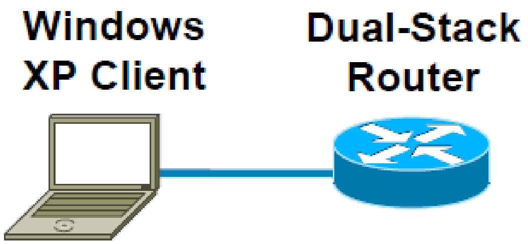
Detail Steps - 2

	Command or Action	Purpose
Step 5	**exit** **Example:** `Router(config-if)# exit`	Exits interface configuration mode, and returns the router to global configuration mode.
Step 6	**ipv6 unicast-routing** **Example:** `Router(config)# ipv6 unicast-routing`	Enables the forwarding of IPv6 unicast datagrams.

Note: “ipv6 unicast-routing” command is a MUST for ipv6 to work in Cisco router

Client Configuration (Windows XP): Dual-Stack (IPv4 + IPv6)

- Required
  - Microsoft Windows XP (SP1 or higher)
  - Microsoft Windows Server 2003
- IPv6 must be installed
  - C:\>ipv6 install
- Have network (routers/switches) configured for IPv6
  - Stateless autoconfiguration and/or DHCPv6




```
C:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 1:
        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 10.1.1.100
        Subnet Mask . . . . . . . . . . . : 255.255.255.0
        IP Address. . . . . . . . . . . . : 2001:DB8:C003:1122:203:ffff:fe81:d6da
        IP Address. . . . . . . . . . . . : fe80::203:ffff:fe81:d6da%4
        Default Gateway . . . . . . . . . : 10.1.1.1
                                            fe80::201:42ff:fe2d:9580
```

Client Configuration (Linux): Dual-Stack (IPv4 + IPv6)

IP address configuration

```
root@box:~# ip addr add 192.168.1.1/24 brd + dev eth1
root@box:~# ip -6 addr add 2001::1/64 dev eth1
root@box:~# ip addr show dev eth1
```

Default gateway IP configuration

```
root@box:~# ip route add default via 192.168.1.254
root@box:~# ip route show
root@box:~# ip route add default via 2001::1000
root@box:~# ip -6 route show
```

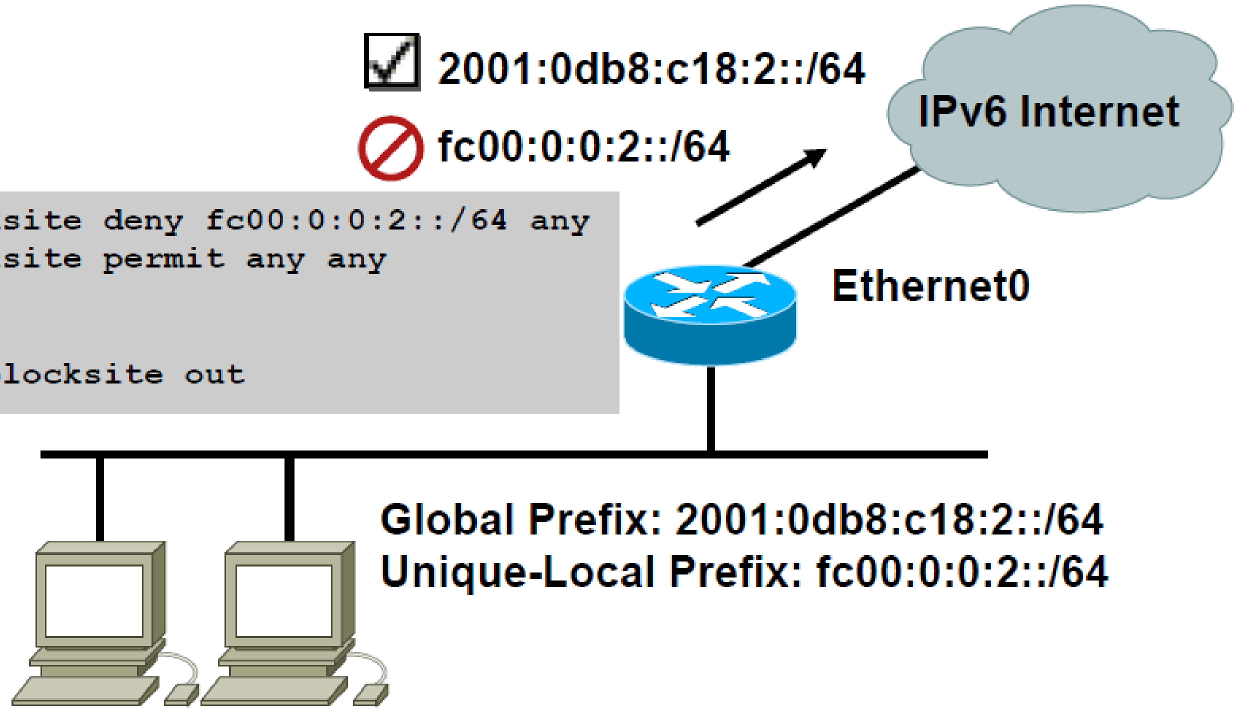
- dev = device
- brd = broadcast
- Unlike Windows, Linux set the IP address and gateway IP separately.

IPv6 Access-List Example

- Filtering outgoing traffic from site-local source addresses

```
ipv6 access-list blocksite deny fc00:0:0:2::/64 any
ipv6 access-list blocksite permit any any

interface Ethernet0
 ipv6 traffic-filter blocksite out
```



IPv6 Access-List Configuration

```
hostname R<No.>
!
ipv6 unicast-routing
!
interface Ethernet0/0
 no ip address
 ipv6 address 3FFE:B00:FFFF:1::/64 eui-64
 ipv6 address 3FFE:B00:FFFF:1001::/64 eui-64
 ipv6 address FEC0:0:0:1::/64 eui-64
 ipv6 enable
 ipv6 nd prefix 3FFE:B00:FFFF:1::/64 300 0
 ipv6 nd prefix 3FFE:B00:FFFF:1001::/64 300 300
 ipv6 nd prefix FEC0:0:0:1::/64 300 300
 ipv6 rip lab4 enable
!
interface Ethernet1/0
 no ip address
 ipv6 address 3FFE:B00:FFFF::/64 eui-64
 ipv6 address FEC0::1/64
 ipv6 enable
 ipv6 traffic-filter lab5 out
 ipv6 rip lab4 enable
!
ipv6 router rip lab4
!
ipv6 access-list lab5
 deny ipv6 FEC0:0:0:1::/64 any
 permit ipv6 any any
```

IPv6 Routing

Default and Static Routing

- Similar to IPv4. Need to define the next hop/interface.
- Default route denoted as ::/0

*ipv6 route ipv6-prefix/**prefix-length {ipv6-address** | **interface-type interface-number [ipv6-address]} [administrative-distance] [administrative-multicast-distance** | unicast | multicast**]** [tag **tag]***

Examples:

- Forward packets for network 2001:DB8::0/32 through 2001:DB8:1:1::1 with an administrative distance of 10

```
Router(config)# ipv6 route 2001:DB8::0/32 2001:DB8:1:1::1 10
```

- Default route to 2001:DB8:1:1::1

```
Router(config)# ipv6 route ::/0 2001:DB8:1:1::1
```

** Administrative distance comes in default values and indicates the reliability of the routing protocol

IPv6 Static Route Configuration

```
hostname R<No.>
!
ipv6 unicast-routing
!
interface Ethernet0/0
 no ip address
 ipv6 address 3FFE:B00:FFFF:1::/64 eui-64
 ipv6 address 3FFE:B00:FFFF:1001::/64 eui-64
 ipv6 address FEC0:0:0:1::/64 eui-64
 ipv6 enable
 ipv6 nd prefix 3FFE:B00:FFFF:1::/64 300 0
 ipv6 nd prefix 3FFE:B00:FFFF:1001::/64 300 300
 ipv6 nd prefix FEC0:0:0:1::/64 300 300
!
interface Ethernet1/0
 no ip address
 ipv6 address 3FFE:B00:FFFF::/64 eui-64
 ipv6 address FEC0::1/64
 ipv6 enable
!
ipv6 route 3FFE:D00:FFFF::/64 FEC0::A
ipv6 route ::/0 FEC0::A
```

RIPng (RFC 2080)

Similar IPv4 features

- Distance vector, radius of 15 hops, split horizon, and poison reverse
- Based on RIPv2

Updated features for IPv6

- IPv6 prefix, next-hop IPv6 address
- Uses the multicast group FF02::9, the all-rip-routers multicast group, as the destination address for RIP updates
- Uses IPv6 for transport
- Named RIPng

Configuring & Verifying RIPng for IPv6

RouterX(config)#

```
ipv6 router rip tag
```

- Creates and enters RIP router configuration mode

RouterX(config-if)#

```
ipv6 rip tag enable
```

- Configures RIP on an interface

```
show ipv6 rip
```

- Displays the status of the various RIP processes

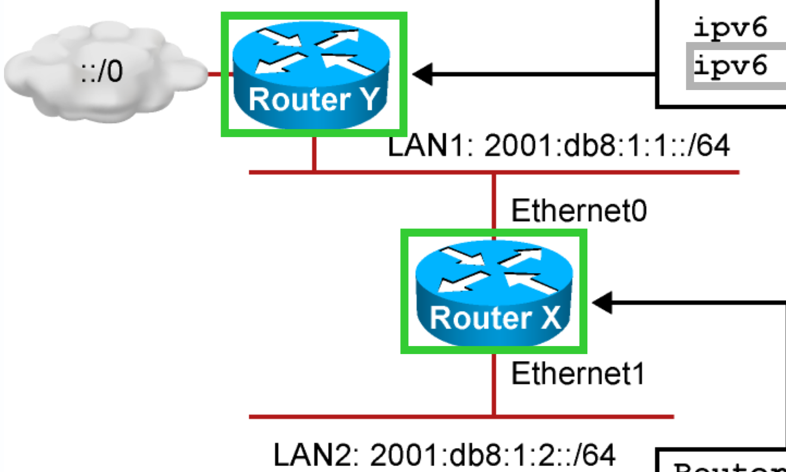
```
show ipv6 route rip
```

- Shows RIP routes in the IPv6 route table

RIPng for IPv6 Configuration Example

```
RouterY#
ipv6 unicast-routing
ipv6 router rip RT0

interface Ethernet0
 ipv6 address 2001:db8:1:1::/64 eui-64
 ipv6 rip RT0 enable
```




```
RouterX#
ipv6 unicast-routing
ipv6 router rip RT0

interface Ethernet0
 ipv6 address 2001:db8:1:1::/64 eui-64
 ipv6 rip RT0 enable

interface Ethernet1
 ipv6 address 2001:db8:1:2::/64 eui-64
 ipv6 rip RT0 enable
```

327P-665

IPv6 RIP Configuration

```
hostname R<No.>
!
ip subnet-zero
no ip domain lookup
!
ipv6 unicast-routing
!
interface Ethernet0/0
 no ip address
 ipv6 address 3FFE:B00:FFFF:1::/64 eui-64
 ipv6 address 3FFE:B00:FFFF:1001::/64 eui-64
 ipv6 address FEC0:0:0:1::/64 eui-64
 ipv6 enable
 ipv6 nd prefix 3FFE:B00:FFFF:1::/64 300 0
 ipv6 nd prefix 3FFE:B00:FFFF:1001::/64 300 300
 ipv6 nd prefix FEC0:0:0:1::/64 300 300
 ipv6 rip lab4 enable
!
interface Ethernet1/0
 no ip address
 ipv6 address 3FFE:B00:FFFF::/64 eui-64
 ipv6 address FEC0::1/64
 ipv6 enable
 ipv6 rip lab4 enable
!
ip classless
!
ipv6 router rip lab4
```

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Quiz Question 1

Which address type from IPv4 was eliminated in IPv6?

a) unicast
b) multicast
c) everycast
d) broadcast

The correct answer is d.

Quiz Question 2

How can you condense consecutive sets of zeros in an IPv6 address?

a) with the “:::” symbol
b) by eliminating leading zeros
c) by replacing four consecutive zeros with a single zero
d) with the “::” symbol

The correct answer is d.